MIL-M-80209B
20 September 1985
SUPERSEDING
MIL-M-80209A
7 March 1975

MILITARY SPECIFICATION

MILLING MACHINE, VERTICAL KNEE TYPE, 3 AXIS, NUMERICALLY CONTROLLED, 1.5 to 7.5 HORSEPOWER SPINDLE DRIVE

This specification approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers numerically controlled, vertical knee type milling machine, with equipment and accessories as described herein.

1.2 Classification. This specification covers machines of the following types, classes, and sizes. The type, class, or size required for a specific procurement shall be as specified (see 6.2.1).

Type I - Numerical control of x, y and quill (z) movement with manual movement of knee (w).

Type II - Numerical control of x, y and knee (w) movement with manual movement of quill (z)

Type III - Numerical control of x, y and knee (z) movement with stationary spindle

Class A (1.5 hp) 8 x 36 inches working table size

Class B (2 hp) 10 x 42 inches working table size

Class C (3 hp)
Size 1 - 9 x 30 inches working table size
Size 2 - 12 x 49 inches working table size

Beneficial comments (recommendations, addition, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3417

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Class D (4 hp) 10 x 50 inches working table size
14 x 47 inches working table size

Class E (5 hp) 9 x 30 inches working table size
11 x 58 inches working table size

Class F (7 and 7.5 hp) 10 x 47 inches working table size
11 x 58 inches working table size

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARD

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURERS ASSOCIATION

AGMA 360.02 - Manual For Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators (DoD Adopted).

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standards for (DoD Adopted).

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues (DoD Adopted).

ANSI B5.18 - Spindle Noses and Tool Shanks for Milling Machines (DoD Adopted).

ANSI B11.8 - Drilling, Milling, and Boring Machines - Safety Requirements for the Construction, Care and Use (DoD Adopted).

(Application for copies should be addressed to the American National Standards Insititute, Department 969, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951 - Commercial Packaging Standard Practice for (DoD Adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, Sales and Service Department, 1916 Race Street, Philadelphia, PA 19103.)

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ELECTRONICS INDUSTRIES ASSOCIATION (EIA)

RS-227 - One Inch Perforated Tape (DoD Adopted).

RS-232 - Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange (DoD Adopted).

RS-244 - Character Code for Numerical Machine Control Perforated Tape (DoD Adopted).

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine Tools (DoD Adopted).

RS-274 - Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines (DoD Adopted).

RS-281 - Electrical Construction Standards - Machine Numerical Control (DoD Adopted).

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape. (DoD Adopted).

RS-494 - 32 Bit Binary CL (BCL) Exchange Input Format for Numerically Controlled Machines.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (IOS)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

ISO/R 1122 - Gears, Glossary of, Geometrical Definitions.

(Application for copies should be addressed to the International Organization For Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Design. The milling machine shall be new and one of the manufacturer's current models. Its design shall include features and components necessary for maintaining alignment and accomplishing drilling, tapping, milling, and boring as required herein. The machine shall be of a fixed base type with vertical movements of the knee on the column ways. Cross movement shall be provided by movement of the table on saddle ways supported by the knee. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376A.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gages, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI system of Measurements. When a dual US and SI system is furnished, both measurment systems shall have independent zero adjustments and both systems shall be calibrated in such a manner that the last graduation progresses into and is continuous with the first graduation as the axis is moved to the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The milling machine and all its applicable components that directly consume energy in normal operation shall be designed

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and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the milling machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 and ANSI B11.8 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservior shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible. In the event of a lube failure, a means shall be provided to automatically inhibit axis movement until the problem is corrected.

3.2.9 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or the machine.

3.3 Construction. The milling machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solder or paste shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of

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these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.001 inch. Dial diameter shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Gears. All gears and pinions used in the machine or its related components shall be properly selected for the intended application and fabricated to either English (USA) standard diametral pitch per AGMA 390.03, AGMA 360.02 or Metric (SI) Module Ratio per the International Organization for Standardization, ISO/R1122, ISO-54, and FED-STD-376A.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

3.4 Components. The milling machine shall consist essentially of a column and base, knee, saddle, worktable, spindle, electric motor drive, and such other components as are required to perform as specified herein.

3.4.1 Column and base. The column and base shall be one-piece or two-piece construction with access doors or removable covers for servicing the drive motor, main drive, clutch and adjusting mechanism. The walls shall be ribbed for rigidity. The column shall have machined ways of the square or dovetail design for vertical travel of the knee. All ways shall be machined

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and scraped or hardened and precision ground for smooth, accurate travel. Means shall be provided for securing the machine to the floor.

3.4.2 Knee. The knee shall be of the box type construction with cross or transverse ribbing. It shall be fitted to the column ways with adjustable gibs and equipped with wipers or other means for preventing dust and chips from entering the ways. Means shall be provided to clamp the knee in any position within its travel on the column. The knee elevating screw shall have either square or acme threads except that when knee movement is numerically controlled, an anti-backlash recirculating ball feed screw shall be furnished. All feed screws shall be suitably protected.

3.4.3 Saddle. All way surfaces of the saddle shall be finished by either scraping or grinding. The ways shall be equipped with adjustable gibs and guides as necessary to maintain alignment in cross and longitudinal planes. Wipers or other means shall be provided to prevent dust and chips from entering the ways.

3.4.4 Table. The table shall be of rigid construction and have not less than three equally spaced T-slots conforming dimensionally with ANSI B5.1. The table working surface shall be the manufacturer's standard size but shall be not less than the size shown in table I, applicable to size and class machine procured. The top of the table shall have a finished surface. The table shall have coolant drain channels with chip screen and provisions for return of coolant to the reservoir. The table and all slides shall be provided with a positive means of securing the machine components in place to permit accurate cuts under full horsepower, without drifting.

3.4.5 Spindle. A variable speed spindle capable of both clockwise and counterclockwise direction of rotation driven by an electric motor shall be provided. The spindle shall be fixed and held in axial and radial alignment by ball or roller bearings. The spindle shall be heat treated, hardened and ground to a precision finish and shall have a taper conforming to ANSI B5.18 unless otherwise specified. The spindle shall be provided with a method for quick tool changes.

3.4.6 Tool changer and storage magazine. When specified (see 6.2.1), the machine shall be equipped with an automatic tool change and storage magazine with a minimum storage capacity of 10 tools. Tool changer shall be accomplished automatically by a power activated arm which will exchange the tool in the spindle with a tool selected from the tool storage magazine. The magazine shall permit convenient loading and unloading of the tool holders during operational setup. Tool selection may be either random or sequential unless specified (see 6.2.1).

3.4.7 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to

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TABLE I. Sizes and capacities. (minimum)

Characteristic	Class A	Class B	Class C		Class D		Class E		Class F	
			Size 1	Size 2	Size 1	Size 2	Size 1	Size 2	Size 1	Size 2
Spindle horsepower	1.5	2	3	3	4	4	5	5	7	7-1/2
Table Size (inches)	8 x 36	10 x 42	9 x 30	12 x 49	10 x 50	14 x 47	9 x 30	11 x 58	10 x 47	11 x 58
Table Travel (inches)										
X	18	18	20	24	26	24	20	32	30	32
Y	11	11	10	14	14.5	14.5	10	14.5	11	14.5
Quill travel (NA to type III Machines)	4"	5"	4.75"	4.75"	5"	4"	6"	6.5"	7"	6.5"
Knee travel	11"	16"	11"	11"	16.5"	15"	11"	16.5"	17"	16.5
Spindle speed range, RPM	260 to 3400	60 to 3300	60 to 3600	100 to 4000	60 to 6000	60 to 3500	60 to 4100	60 to 4100	100 to 4675	60 to 4100
Feedrate range X and Y axes	0 to 200	.1 to 100	0 to 100	0 to 100	.1 to 100	.1 to 100	.3 to 100	.3 to 100	.1 to 100	.1 to 100
Z axis (IPM)	0 to 200	—	0 to 100	0 to 100	.1 to 100	.1 to 100	.3 to 100	.3 to 100	.1 to 100	.1 to 100
Power Rapid Traverse/IPM										
X and Y axes	300	200	250	250	300	200	200	200	200	200
Z axis	200	200	250	250	300	200	200	200	200	200

NOTE: Machine size and capacity shall be equal to or greater than the stated requirements and less than the requirements for the next larger size as shown in above table. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater. If the largest size covered in the above table is ordered, the maximum size and capacity acceptable shall not exceed the stated requirements by more than 10 percent.

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permit the operator to direct and control the amount of coolant. A splash-back shall be provided to prevent coolant and chips from being lodged between the column and the electrical cabinets. A base or floor mounted splash tray shall be provided to catch coolant overflow from the chip and coolant channels around the working surface of the table.

3.4.8 Hydraulic system. The hydraulic system, when furnished, shall be complete, including all pumps, valves, piping, cylinders and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.9 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA-79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA-79 for the disconnect fuse rating.

3.4.9.1 Motors. Motors shall be rated for a short duty cycle of not less than 60 minutes and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall be in accordance with ANSI/NEMA MG-1.

3.4.10 Solid state components. Electrical devices used to power electric servo and spindle motors or control voltage delivered to solid state electrical components shall be solid state devices. Such devices shall be capable of handling the electrical load imposed by operation of the machine at its maximum capability. The use of selinium in solid state components shall be restricted to devices that protect other solid state components from voltage surge.

3.4.11 Computer numerical control (CNC). Unless otherwise specified (see 6.2.1), the milling machine will be equipped with a fully automatic solid state soft-wired, integrated circuit type numerical control. The numerical control shall provide automatic control of machine functions, operating modes, spindle speeds, direction of rotation and simultaneous control of the machine's three linear axes in linear and helical interpolation. The control shall also be provided with circular interpolation capabilities. Machine axes shall be identified in accordance with EIA-RS-267. Line voltage fluctuation of +10 percent from the normal operating voltage shall not adversely affect the operation of the CNC. Unless otherwise specified (see 6.2.1), the control shall be provided for connection to a 115 volt AC 60 HZ, single phase power source. The control shall be capable of operating in an

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outside ambient temperature of 50-110 degrees F at 5-95 percent non-condensating relative humidity. The control shall be automatically shut down when the maximum operating temperature is reached. Program resolution shall be 0.0001 inch or 0.001 mm for each linear axis. The mechanical and electrical design of the control shall comply with EIA-RS-281 construction standards. The control shall be capable of accepting input data in the form and format specified by the ordering activity.

3.4.11.1 Punched tape data input. Unless otherwise specified (see 6.2.1), the input media for the control shall be one inch, eight channel punched tape complying with EIA-RS-227.

3.4.11.1.1 Tape reader. When one-inch tape is the specified input media, the control shall be equipped with a photoelectric tape reader capable of holding, feeding and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2.1), the tape reader shall be supplied with standard feed and take-up spoolers 5-1/4 inches in diameter. The tape reader shall be housed in a dust proof enclosure mounted in a control cabinet and shall be easily accessible for cleaning and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244
b. Even parity per EIA RS-358
c. Misaligned sprocket holes
d. No sprocket holes

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.11.2 Flexible diskette data input. When 32 Bit Binary Cutter Location (BCL) Exchange Input for numerically controlled machines is specified (see 6.2.1), the input media shall be an eight inch flexible diskette recorded in accordance with 3.4.11.2.1. The input data format shall comply with EIA Standard RS-494. Manufacturers providing functions or features that are not currently covered in EIA-RS-494 shall apply to the Electronic Industries Association, IE-31 Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the manufacturer's features or functions. The control shall be equipped with an eight-inch flexible diskette reader. The reader shall have the capability to perform a cyclic redundancy check of the input data and be capable of reading not less than 15,000 characters per second.

3.4.11.2.1 Flexible diskette format. When an eight-inch flexible disk is specified as the input media to the numerical control it shall be a single density recording on one side using a soft sectored format. The disk shall be divided into 77 distinct locations on the single-side disk. Tracks shall be assigned numbers from 00 through 76. Track 00 shall be nearest to the edge of the disk and track 76 nearest the center. Tracks 00, 74, 75 and 76 shall not be used. There is an index hole in the jacket and the disk that

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shall be used for timing functions when a beam of light passes through it. Recording shall be accomplished by rotating the disk inside its jacket exposing the disk at the read/write head slot cut in the jacket. Data shall be recorded only on the side of the disk that is exposed to the read/write head. Each track shall be divided into 26 addressable sectors which contain 128 bits. All sectors shall be identified by the file starting sector number with track 00 sector 1, designated as 0. The first part program shall be located at sector 1 E hexidecimal corresponding to track 1, sector 5. Where two byte numbers are referenced the high order byte shall be recorded first and the low order last. Track 1, sectors 1 through 4 shall contain the program directory of up to 16 part program names in ASCII. Each file shall be packed on the disk in the same order as its directory entry. Unused directory entries shall point to the next available sector after the last file and contain the 30-character ASCII space characters. The last sector of each file shall be padded with end of file (EOF) records (8000 0000). Diskette addressing shall comply with the requirements of table II.

3.4.11.3 Program features.

3.4.11.3.1 Tape coding. The control shall accept punched tape programmed in accordance with EIA-RS-274. The control shall be capable of recognizing and automatically accepting coding in accordance with EIA-RS-244 or EIA-RS-358.

3.4.11.3.2 Leading zero suppression. The control shall be capable of suppressing leading zeros.

3.4.11.3.3 Selectable inch/metric input. The control system shall provide operator or program selectable inch/metric input capability.

3.4.11.3.4 Absolute/incremental input. The control shall have switchable absolute/incremental input capability.

3.4.11.3.5 Departure control. The maximum departure permitted by a single command in either incremental or absolute mode shall be not less than plus or minus 99.9999 inches or 999.999 mm. The control shall be capable of automatically controlling the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall not be more than 0.0001 inch or 0.001 mm.

3.4.11.3.6 Preparatory function (G-Codes). The preparatory function codes of the control shall be in accordance with EIA RS-274.

3.4.11.3.7 Miscellaneous functions (M-Codes). The miscellaneous function codes of the control shall be in accordance with EIA RS-274.

3.4.11.3.8 Additional features. Unless otherwise specified (see 6.2.1), the following additional numerical control features shall be provided:

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TABLE II. Diskette addressing.

Sector Address	* Track	* Sector Number	Byte	Use
0 - 25	0	All	All	Unused
20 - 29	1	1 - 4	0 - 1	File start sector (binary)
			2 - 31	30-character ASCII program name to be defined by user
			32 - 63	Same as 2 - 31 above for next part program (file)
			64 - 95	Same as 32 - 63 above for next part program
			96 - 127	Same as 32 - 63 above for next part program
		5 - 26	All	APT CL records for NC part programs
30 -1897	2 - 73	All	All	APT CL records for NC part programs
1898 - 1975	74 - 76	All	All	Unused

* This numbering corresponds to IBM 3740 physical format.

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a. Four quadrant programming.
b. Incremental and continuous jog for each axis.
c. Inches per minute (IPM)/Millimeter Per Minute (MMPM).
d. Vector feedrate.

3.4.11.4 Axis and tool management.

3.4.11.4.1 Tool length offsets. The control shall have a tool length offset feature to permit adjustments to the programmed tool length by means of manual data input. The tool length offset feature shall have a range of not less than the following:

a. Inch system ±0.0001" to ±1.0000".
b. Metric system ±0.001 mm to ±25.400 mm.

3.4.11.4.2 Cutter compensation. Cutter compensation shall be provided to compensate for cutter diameters differing from those utilized in programming. The cutter compensation shall have a range of not less than the following:

a. Inch system ±0.0001" to ±1.0000".
b. Metric system ±0.001 mm to ±25.400 mm.

3.4.11.4.3 Fixture compensation. A fixture compensation feature shall be provided to permit compensation for fixture position errors.

3.4.11.4.4 Reversal error compensation. Reversal error compensation circuitry shall be provided for a selectable dimension correction to be added automatically to the command dimension to each linear axis reversal to compensate for mechnical lost motion.

3.4.11.4.5 Lead error compensation. Lead error compensation software shall be provided to permit compensating for the lead errors of each axis ballscrew.

3.4.11.4.6 Auxiliary functions. Unless otherwise specified (see 6.2.1), the MCU shall respond to the following program command auxiliary functions: interpolation, fixed cycles, and dwell. Unless otherwise specified (see 6.2.1), miscellaneous function shall include program stop, end of program, spindle CW, spindle CCW, spindle off, coolant on-off tool change, end of program (rewind), and program re-start. System preparatory and miscellaneous functions shall be coded in accordance with EIA RS-274.

3.4.11.5 Operator interface and controls.

3.4.11.5.1 Data display. A cathode ray tube (CRT) capable of displaying at least 250 characters simultaneously, shall be provided. Unless otherwise specified (see 6.2.1), the display shall, as a minimum, display the following data:

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a. Four digit block sequence number.
b. Active block of stored data.
c. Actual position of each axis (may be a LED display on the control unit).
d. Control and system error messages.

3.4.11.5.2 Control system modes of operation. The operational modes of the NC control shall include manual, automatic, single block, MDI, and sequence number search.

3.4.11.5.3 Manual data input. A manual data input (MDI) unit shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.11.5.4 Block delete. The control shall have a block delete feature to permit by-passing of blocks of programmed data.

3.4.11.5.5 Block-by-block read. The control shall have block-by-block read function to provide for reading information one block at a time.

3.4.11.5.6 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequenced number.

3.4.11.5.7 Part program edit. A program edit capability shall be provided to allow block insertion, deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.11.5.8 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent.

3.4.11.5.9 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment or in steps of not greater than 10 percent each. The spindle speed override control shall control not less than the range of override percentages between 60 and 120 percent of programmed speeds.

3.4.11.5.10 Mirror image. The control shall be capable of producing either right or left-hand parts from the same input data through the inversion of x and y axis. Inversion of each axis shall be by keyboard by switch or by programmed entry. Mirror image shall not affect the directional sense of manual controls. Reverse settings shall be displayed on the data display.

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3.4.11.5.11 Axis jog. The control panel or pendant shall have manual controls for accomplishing both single and continuous, low and high speed jog movements of the slides for each controlled axis in both the plus and minus direction.

3.4.11.5.12 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.11.5.13 Program tryout. The control shall have a program tryout feature to permit new part programs to be read through the control without machine motions.

3.4.11.5.14 Operator control panel. The control unit shall have a control panel which provides, unless otherwise specified (see 6.2.1), at least the following control functions: Emergency Stop; Cycle Start; Optional Stop; Mode Selection; Feed Hold; Feedrate Override; Spindle Speed Override; Jog; Jog Direction; and EOB Stop.

3.4.11.6 Miscellaneous control requirements.

3.4.11.6.1 Part program storage. Part program memory loadable from the data input media shall be provided. Unless otherwise specified (see 6.2.1), the part program memory shall be capable of storing not less than 100 feet of postprocessed part program tape.

3.4.11.6.2 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory, to hold the size memory provided at full load for a period of not less than 72 hours when the primary power source fails.

3.4.11.6.3 Operational software. Operational software shall be resident in the control and shall contain all logic necessary to effectively operate the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the controls, machine tool, part program, and operator's actions.

3.4.11.6.4 Programmable interface. A software program for accommodating the softwired interface between the numerical control and machine shall be provided as part of the executive program or as a separate program.

3.4.11.6.5 Maintenance diagnostic systems. Maintenance diagnostics software shall be furnished in either the control as non-volatile memory or in the form of 1-inch tape or 8-inch flexible disk, as required by the specified input media, to provide for diagnostic checking of the machine control unit. The diagnostic software shall test, exercise, and display failures, at least to board level.

3.4.11.6.6 Peripheral equipment interface. The control shall be equipped with an interface complying with EIA RS-232.

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3.4.11.6.7 Canned cycles. Unless othewise specified (see 6.2.1), the following canned cycles shall be provided:

a. Drilling
b. Boring
c. Tapping
d. Bolt hole circle

Such cycles shall be called by G codes and shall be initiated when X, Y, Z and R plane data is entered plus data for the F, E and K words.

3.4.11.6.8 Operator portable or pendant control. When specified (see 6.2.1), an operator's control of the pendant or portable box type shall be provided for the CNC with features as specified by the procuring activity.

3.4.11.6.9 Postprocessor. Unless otherwise specified (see 6.2.1), the CNC shall be furnished with postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be provided for converting specified part program language (see 6.2.1) to the form and format required for input to the CNC. The postprocessor shall be written for use with the computer specified (see 6.2.1). Documentation sufficient to implement and use the postprocessor shall be provided for computer compatability verification and validation at least 30 days prior to delivery of the milling machine.

3.4.11.6.10 Conversational programming capability. When specified by the procuring activity conversational programming capabilities shall be provided (see 6.2.1).

3.4.11.6.11 Graphic display capability. When specified by the procuring activity, graphic display capabilities shall be provided (see 6.2.1).

3.4.11.6.12 Additional data input media. When specified, a secondary input device shall be provided to input data on either one inch eight channel punched tape, magnetic tape cassette or flexible diskette (see 6.2.1). When one inch eight channel punched tape reader is specified the tape reader and control shall satisfy the requirements in 3.4.11.1.1.

3.4.11 6.13 Reader/recorder. When specified (see 6.2.1), the control shall be equipped with a reader/recorder for reading and recording either tape cassettes or flexible disks. When specified (see 6.2.1), a portable tape punch shall be supplied for punching one inch eight channel tape. Recording shall be accomplished when the control is operated in the program record mode.

3.5 Sizes and capacities. The sizes and capacities of the milling machine shall conform to the requirements of table I.

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3.6 Performance. The milling machine and numerical control system shall be capable of satisfactory performance when tested in accordance with 4.5 through 4.6.3.4.

3.7 Alignment tolerances. Each milling machine shall meet the alignment accuracies of table III.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.
b. Operational and programming manuals.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.10 Additional performance requirements. Performance features required in addition to or in lieu of the requirements herein shall be as specified (see 6.2.1).

3.11 Marking of instruction plates, indicating plates and nameplate. All words on instruction plates, indicating plates and name plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background. When these methods are not practicable, the instructions shall be applied directly on the item by environmentally protected decalcomania transfer, metal wrap-around tags, stencil, silk screen or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item to which it is applied and capable of withstanding the environmental tests and cleaning procedures specified for the item.

3.11.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each milling machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each milling machine. The nameplate shall contain the information listed below. If the milling machine is a special model, the model designation shall include the mode of the basic standard mill and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

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TABLE III. Alignment accuracies.

Component Characteristics	Permissible Tolerances
Worktable:	
Flatness of work mounting surface	0.0005"/FT
Rise and fall of table in X-axis	0.0006" in 18"
Rise and fall of table in Y-axis	0.0006" in 9"
Center T-slot parallel with X-axis	0.0006"/FT
Center T-slot square with Y-axis	0.0006"/FT
Worktable square to spindle in X-axis (0° to 180° method; table rising toward the front side only)	0.0008" per 12" dia. Sweep
Worktable square to spindle in Y-axis (90° to 270° method)	0.0008" per 12" dia. Sweep
X-axis travel perpendicular to Y-axis	0.0006"/FT
Knee travel perpendicular to Y-axis	0.0008"/FT
Knee travel perpendicular to X-axis	0.0008"/FT
Spindle:	
Spindle axial runout	0.0002" TIR
Spindle internal taper runout:	
1" from spindle nose	0.0003" TIR
6" from spindle nose	0.0006" TIR
Positioning accuracy:	
(Without corrections or compensation)	
X-axis positioning 1/	±0.0006"/FT
Y-axis positioning 1/	±0.0006"/FT
Z-axis positioning	±0.0004"
Position repeatability	±0.0004"
Lost motion X and Y; axes centered	0.0005" max

NOTE: 1/ Not to exceed 0.001" in overall travel.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching the maximum accumulative hours, the meter readout shall automatically revert to zero and continue to totalize time.

3.13 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of the contract (see 6.2.2).

3.14 Workmanship. Workmanship of the milling machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first item submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

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4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.5 and the tests in 4.6.2. Failure of the item to pass the examination or tests shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.5 and 3.8 through 3.14.

4.6 Tests.

4.6.1 Operational test. The milling machine shall be operated at no load for not less than 3 hours. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.6.2 Alignment accuracy tests. Each milling machine and its numerical control system shall be tested to determine that it complies with the alignment accuracy requirements of table III (see 3.7).

4.6.3 Performance test.

4.6.3.1 Axis positioning tests. Starting at positions other than the extremes of axis travel, two identical movements of each controllable linear axis shall be programmed in each direction. Programmed span shall be not less than 7 inches, and each digit of starting and end point shall be other than zero with respect to reference zero. Feed rate of axis shall be two inches per minute. Absolute positioning error shall be $\pm$0.0006"/ft or less. Similar movements of each linear axis shall be programmed at feed rates nearest to 10, 20 and 100 percent of maximum linear feed rate. Overshoot shall not exceed 0.003 inch at any feed rate up to 20 percent of the maximum. Maximum overshoot shall not exceed 0.005 inch. Stops shall be programmed appropriately for inspection of control positioning accuracy. Positioning tests may be part of the same tape program as 4.6.1.

4.6.3.2 Maximum horsepower test. Using a workpiece of SAE 1020 steel, a straight cut of 12 inches or more shall be made with a face mill or shell end mill. Diameter of the cutter and spindle speed shall be determined by the supplier to be suitable for making the cut with specified chip load and peripheral cutter speeds. The depth of the cut shall be as required to load the spindle motor to its maximum rated horsepower. Chip load shall be 0.010" $\pm$ 0.002" and peripheral cutter speed shall not exceed 300 feet per minute. The cutting action shall be smooth and even and the finished workpiece shall show no evidence of tool chatter.

4.6.3.3 Machining accuracy tests. Using one or more test setups devised by the supplier, tests shall be made to demonstrate the cutting accuracy of the machine under numerical control. The test piece may be, at the supplier's option, cast iron or aluminum, and the cutting tool, spindle speed,

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and feed rate for each test shall be selected by the supplier as suitable for use for the cut to be made. The tests shall be as follows: tolerances shown in 4.6.3.3.1 through 4.6.3.3.3 shall be applicable. The workpiece shall present a work surface normal to the tool point and approximately centered with reference to the work area in the XY plane. Roughing cuts may be made in the various cuts with excess metal reprogrammed.

4.6.3.3.1 Straight line milling tests. The straight line milling test shall be a programmed square or rectangular not less than 6 inches per side with sides parallel to the travel along X and Y axes. A plunge cut shall be made along the longitudinal axis. Plunge angles shall be not less than 2 degrees. Precision tolerances shall be as follows:

Squareness of sides	0.002" total deviation
Parallelism of opposing sides	0.001" total deviation
Length of sides, deviation from programmed length, plus or minus	0.001"
Corner location, deviation from programmed point, plus or minus	0.001"
Plunge cut, rising cut, where applicable, maximum total deviation over span	0.002"

4.6.3.3.2 Circle contouring test. A circle shall be programmed within the outline of the square cut under the provisions of 4.6.3.3.1. Cut shall be made at constant Z-axis depth using the shortest programming sequence within the control capability. Precision tolerances shall be:

Roundness:

12 indicator readings at 30 degree intervals, deviation not more than 0.002 inch from programmed diameter

4.6.3.3.3 Straight line contouring test. A square or an equilateral triangle shall be milled within the outline of the circle described in 4.6.3.3.2 at constant depth of not less than one-half inch. The reference line starting the square or triangle shall be 10 to 20 degrees away from the direction of longitudinal movement (X-axis).

Precision tolerances shall be:

Surface flatness of sides, plus or minus deviation	0.001"
Angularity of sides, plus or minus	0.0015" total

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4.6.3.4 Tapping tests. Tapping tests shall be made under numerical controls as follows. Pilot drill and tap at programmed locations four holes each with finished threads to accept 8-32 and 3/4-10 screws. Finished holes shall accept standard 8-32 or 3/4-10 screws freely and threads shall meet class 3 standards for internal threads as set forth in FED-STD-H28.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified (see 6.2.1), level A packaging, level A or level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The numerically controlled machines covered by this specification are intended for use in any production shop where drilling, boring, tapping, reaming and milling operations by numerical control of the machine are required.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class and size required (see 1.2).

c. When first article is required for inspection and approval (see 3.1).

d. If machine is required to be configured in a specified measurement system (US or SI), state required system (US or SI), state required system (see 3.2..1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Painting, if different (see 3.3.5).

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i. Electromagnetic interference control, if required (see 3.3.9); specify class and requirement.

j. Automatic tool changer, if required (see 3.4.6).

k. Specify tool selection as sequential or random if only one is acceptable (see 3.4.6).

l. Electrical system and input power supply if different (see 3.4.9).

m. Motor enclosure, if different (see 3.4.9.1).

n. Computer numerical control if different (see 3.4.11).

o. Input media, if different; specify rquirement (see 3.4.11.1)

p. Tape reader spools, if different; specify requirement (see 3.4.11.1.1.

q. Thirty-two bit binary cutter location (BCL), if required (see 3.4.11.2).

r. Additional features, if different; specify requirements (see 3.4.11.3.8).

s. Auxillary functions, if different; specify requirements (see 3.4.11.4.6).

t. Data display, if different; specify requirement (see 3.4.11.5.1).

u. Operator control panel, if different; specify requirement (see 3.4.11.5.14).

v. Part program storage, if different; specify requirement (see 3.4.11.6.1).

w. Standby battery power, if different; specify requirement (see 3.4.11.6.2).

x. Canned cycles, if different; specify requirements (see 3.4.11.6.7).

y. Operator portable or pendant control, if required, specify, and the controlled functions and features required (see 3.4.11.6.8).

z. Postprocessor requirements, if different; specify requirements (see 3.4.11.6.9).

aa. Part program language used; specify (see 3.4.11.6.9).

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bb. Computer used for postprocessing; specify (see 3.4.11.6.9).

cc. Conversational programming capability; specify requirements (see 3.4.11.6.10)

dd. Graphic display capability; specify requirements (see 3.4.11.6.11).

ee. Additional input device, if required; specify media (see 3.4.11.6.12).

ff. Reader/recorder for tape cassette or flexible disk; if required, specify which (see 3.4.11.6.13). Portable tape punch; if required, specify (see 3.4.11.6.13).

gg. Standard equipment, if different (see 3.8).

hh. Optional equipment, if required; fully describe (see 3.9).

ii. Specify additional performance features required (see 3.10).

jj. Lubrication chart or plate, if different (see 3.11.1).

kk. Nameplate, if different (see 3.11.2).

ll. Hour meter, if different (see 3.12).

mm. First article inspection, if different (see 4.3).

nn. Quality conformance inspection, if different (see 4.4).

oo. Level of packaging, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. All data requirements, such as parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results shall be required on a DD Form 1423, Contract Data Requirements List, and shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since

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OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the requesting activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of-operation guarding normally furnished as standard on the manufacturer's commercial market. If specific point-of-operation guarding is required, requesting activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Army - AL
Navy - SH
Air Force - 84
DLA - GS

User Activities:
Army - ME
Navy - MC

Preparing Activity:
DLA-IP

Project Number:
3417-0158

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-80209B	MIL-M-80209B, Milling Machine, Vertical Knee Type, 3 Axis, Numerically Controlled, 1.5 to 7.5 Horsepower Spind

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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